Street-level heat mapped by bicycle-mounted sensors

A. Demo and B. Example

Abstract

Cities are warmer than their surroundings, but the difference varies from street to street. We mounted temperature sensors on twelve shared bicycles and mapped air temperature across a mid-sized city during one summer, from June to August 2025. Tree-lined streets were 1.4 degrees cooler than streets without trees at midday.

1 Introduction

Urban heat islands raise heat stress and energy demand. Fixed weather stations are sparse, so the variation between streets is poorly known. Mobile sensors can fill the gaps between stations.

Earlier mobile campaigns used cars or pedestrians. Bicycles cover more ground than pedestrians and reach streets that cars avoid. Bicycles also move slowly enough for a shielded sensor to follow changes in air temperature.

2 Methods

2.1 Sensors

Each bicycle carried a temperature sensor in a ventilated shield, mounted under the saddle. Readings were taken every ten seconds and tagged with the bicycle's position. Before the campaign, all twelve sensors were compared side by side with a reference thermometer for three days; they agreed within 0.2 degrees.

2.2 Processing

Readings were averaged on a grid of 100 m cells. Cells with fewer than five readings were left out. Readings taken while a bicycle stood still for more than a minute were removed, because the shield then heats up in the sun.

3 Results

Over the summer the bicycles recorded 1.9 million readings in 2,140 cells. Tree-lined streets were on average 1.4 degrees cooler than streets without trees at midday.

The difference was largest in the afternoon and smallest at night.

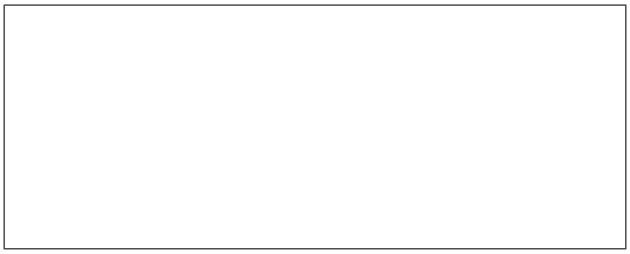


Figure 1: Midday air temperature by grid cell, with the twelve bicycles' routes in grey.

4 Discussion

Shared bicycles are a cheap way to map street-level heat. Planting trees on the warmest streets would reduce heat stress where people walk and cycle. One summer in one city cannot show how the pattern changes between years or climates; a second season is planned.